
Why were 18 of 30 orders late?
Monday Orders (30 total)
1 2 3 4 5 6
7 8 9 10 11 12
13 14 15 16 17 18
19 20 21 22 23 24
25 26 27 28 29 30
Late (18)
On time (12)
Order Process (timings unknown)
Pick
Scan
Pack
Courier
Scanner Charging Checklist
Date
Scanner 1
Scanner 2
Mon
Tue
Wed

What problem are we solving?

3 hours: 50-minute lesson, 10-minute break, 50-minute guided practice, 10-minute break, 50-minute applied practice, and 10-minute wrap-up.

SITUATION

The problem

A small warehouse shipped 18 of 30 orders late on Monday. Staff report that two barcode scanners ran out of battery, but the manager does not yet know whether this explains every delay.

TARGET

Learning goal

Map a simple process, distinguish evidence from assumptions, and propose a small test of a likely process cause.

Practice material: Use the practice material supplied in the session Markdown.

Use these supplied facts

Do not replace missing information with guesses.

CASE MATERIAL

Practice facts

- Orders are picked, scanned, packed, then handed to a courier. Two scanners were uncharged at opening. The charging checklist was blank for three days. No timing data by step is available.

BOUNDARY

Use the supplied facts

A small warehouse shipped 18 of 30 orders late on Monday. Staff report that two barcode scanners ran out of battery, but the manager does not yet know whether this explains every delay.

Understand the idea and apply it

Work through the explanation before comparing prompts or starting the exercise.

LEARN

Describe the process problem with a count

- The supplied observation is that 18 of 30 Monday orders were late. That is a rate of 60% for that sample. It describes the scale of the problem on Monday; it does not show what caused the delays or prove that other days had the same rate. Keep the time period and denominator attached to the number.

LEARN

Use cause questions as hypotheses, not conclusions

- Two scanners were found uncharged and the charging checklist was blank for three days. These facts suggest a possible equipment-readiness issue, but they do not prove it caused all 18 delays. A simple "why" chain helps the team ask what happened earlier in the process. Each proposed link should be marked observed, supported, or still unknown. Do not turn a plausible story into an established root cause.

Understand the idea and apply it

Work through the explanation before comparing prompts or starting the exercise.

LEARN

Test one manageable cause

- If the team suspects scanner readiness, use a one-week charging checklist and record whether scanners are ready at shift start, orders picked per hour, and late orders. Compare the same measures with a recent baseline if available. If scanner readiness improves but lateness does not, weaken or revise the hypothesis. The purpose is to learn which process change helps, not to assign blame.

APPLY

Three-step method

- State the measurable problem; build a short cause chain and label evidence versus unknowns; choose a low-risk test and a measure.

What makes a prompt weak?

These examples come from the current session scenario.

AVOID

Weak example 1

```
Who caused all these late orders?
```

AVOID

Weak example 2

```
Do five whys and tell me the root cause.
```

A clearer, safer prompt

Keep the source, task, constraints, and output check visible.

USE

Improved example 1

```
Use only the facts below. List what is observed,
what is unknown,
and up to three possible causes of late orders.
Mark each cause as a hypothesis; do not assign
blame.
```

USE

Improved example 2

```
For the hypothesis that uncharged scanners slowed
picking, propose a one-week test.
Specify the daily measure, comparison, and a
result that would weaken this hypothesis.
Do not assume it explains all 18 late orders.
```

Exercise 1: Follow the steps

Use the complete case and instructions from the session Markdown.

GUIDED PRACTICE

Exercise 1: Follow the steps

- Ask AI to list evidence, unknowns, and possible causes. Draw a simple chain from uncharged scanners to a possible picking delay. Mark every unverified link with "to check."

Exercise 2: Try a small variation

Use the complete case and instructions from the session Markdown.

APPLIED PRACTICE

Exercise 2: Try a small variation

- Draft a one-week opening checklist test. Record scanner charge status and late orders daily; compare with the previous week if reliable records exist. State why the result can suggest but not prove causation.

Write your session report

Use Report_template.docx in Microsoft Word. Save it as FirstName_StudentID_W10_Keyword.docx and upload it to iLearn by 23:59 on class day.

SUBMIT

One report for this session

- Introduction: Describe the warehouse's late orders and what the manager does not yet know.
- Method: Copy the prompt you used to list evidence, unknowns, and possible causes, and the prompt you used for the test plan.
- Results: Include the cause map with every unverified link marked "to check."
- Conclusion: Give the one-week test plan with its measure, time period, and one limitation.

How to know the work is complete

Check your report before you hand it in.

VERIFY

Success check

- The response separates observations from hypotheses, avoids blaming workers, and defines a measurable test.